

Self-Aware High-Sensitivity Clot Monitoring: A Clinical-Grade Bayesian Triple-Encoder Hybrid Transformer Architecture for Wearables

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Abstract—Post-surgical patients face serious clot risk, yet current monitoring only reacts once symptoms surface. This paper introduces the SWAT (Spatial-Wave-Attention-Temporal) Tier, a wristband-based thrombosis detection system built around a Bayesian triple-encoder architecture joining a 1D-CNN, a multi-head Transformer, and a stacked Bidirectional LSTM. Each encoder was chosen for a specific physiological reason tied directly to Virchow’s Triad of clot formation rather than architectural convention.

The training corpus was acutely lopsided: just 44 of 7,820 labelled windows fell into the Critical risk class. A Wasserstein GAN with Gradient Penalty (WGAN-GP) grew that pool to 2,400 statistically faithful synthetic examples, making genuine discriminative learning feasible. Fifty stochastic MC-Dropout forward passes then score every prediction by Shannon Entropy, routing uncertain readings to a human-review flag rather than a clinical alert. A custom decision gate ($\tau = 0.109$) replaces the standard argmax classifier, deliberately trading some precision for recall because a missed clot is categorically worse than an unnecessary nursing check.

On six post-operative patients the model had never seen during training, the system logged 86.0% Critical Recall and an F1-score of 0.90, while flagging the onset of a confirmed thrombotic event 3 hours and 12 minutes before clinical ultrasound could confirm it. After INT8 quantisation the model fits in 0.66MB and runs in 2.1ms on an ARM Cortex-M7 microcontroller—a 54% size reduction with under 1% loss in recall. A full ablation study confirms that every component earns its place.

Index Terms—Thrombosis monitoring, Bayesian Transformers, WGAN-GP, clinical AI, wearable sensors, multi-modal fusion, epistemic uncertainty, Shannon entropy, Virchow’s Triad, edge inference, INT8 quantisation, MC-Dropout.

I. INTRODUCTION

A. Why Blood Clots Demand a Better Monitoring Strategy

Deep Vein Thrombosis and Pulmonary Embolism together constitute venous thromboembolism (VTE), affecting roughly 1–2 per 1,000 adults each year and ranking as the third most frequent acute cardiovascular syndrome globally, behind only heart attack and stroke [16]. What makes it so clinically treacherous is that it typically builds silently: a clot can grow and migrate for hours before producing any noticeable sign. By the time limb swelling or breathing difficulty appears, the window for low-risk intervention may already be closing.

Standard diagnostic tools—duplex ultrasound and MR venography—are ordered reactively, triggered by visible clinical signs. That approach works for the average patient but is fundamentally misaligned with what post-surgical, oncological, and long-term immobile patients actually need, since these groups carry the highest documented VTE risk and require surveillance that does not wait for symptoms to appear.

B. What Makes Wearable Thrombosis Detection Hard

Wristband sensors offer a route to continuous, non-invasive monitoring, but translating that potential into clinical practice runs into three stubborn problems.

Radically imbalanced training data. Over the entire 14-month data collection described in this paper, only 44 of 7,820 labelled 30-second windows fell into the Critical risk class—roughly 1 in every 178 samples. Standard deep learning optimisers minimise overall loss, which means they learn, quite rationally, to almost never predict Critical. Global accuracy stays high while the model becomes clinically useless for the

one event that actually matters. We call this the **Accuracy Paradox**.

Physiology is personal. A resting heart rate of 58bpm is normal for one patient and near-bradycardia for another. Skin conductance varies enormously by individual. A classifier trained with fixed thresholds on one population misfires constantly on a different one, and no standard argmax output tells the clinician how much to trust any given prediction.

Motion breaks the signal. A patient reaching for a glass of water produces a BVP artefact that can look, numerically, like acute haemodynamic stress. If the model cannot separate motion noise from real physiology, it will generate enough false alarms that nursing staff stop trusting it within days of deployment.

C. What This Paper Contributes

The **SWAT Tier** resolves all three problems through four concrete innovations:

- 1) A tri-encoder design (CNN–Transformer–BiLSTM) where each component was chosen for a specific physiological reason tied to Virchow’s Triad.
- 2) A WGAN-GP augmentation pipeline that builds a realistic Critical-class manifold from just 44 real examples, making discriminative boundary learning possible where it was previously not.
- 3) MC-Dropout uncertainty scoring with a Shannon Entropy gate that causes the model to abstain from ambiguous predictions and request a cleaner reading rather than guessing.
- 4) An asymmetric decision threshold ($\tau = 0.109$) calibrated to the real-world cost structure, where a missed clot is catastrophic and a false alarm is merely inconvenient.
- 5) INT8 post-training quantisation confirmed on ARM Cortex-M7 hardware, demonstrating that clinical-grade inference can run on a wristband chip.

II. RELATED WORK

A. How Clots Are Currently Diagnosed

Duplex ultrasound reaches roughly 95% sensitivity for proximal DVT but requires a trained sonographer and cannot run continuously [16]. The D-dimer blood test is highly sensitive but nearly useless for specificity (40–60%) in post-surgical patients, since surgery itself elevates coagulation markers in almost everyone and fills the result sheet with false positives. Portable Doppler devices improve bedside access without solving the fundamental problem: the interval between consecutive tests is precisely when asymptomatic thrombus propagation happens.

B. What Commercial Wearables Can and Cannot Do

The Apple Watch Series 9 and Fitbit Sense 2 have made atrial fibrillation detection routine. Both target rapid electrical

anomalies in the cardiac cycle where a single 30-second PPG recording can be diagnostic. Thrombosis is slower and multi-modal: venous stasis builds over hours, peripheral temperature regulation shifts gradually, and the autonomic response unfolds across minutes rather than milliseconds. No current commercial platform captures those dynamics or monitors for the progressive damping of distal pulse wave amplitude that signals compromised venous return.

Xu *et al.* [11] showed that on-device biosignal classification can achieve sub-5 ms latency through INT8 quantisation, which settled the hardware feasibility question. Their work did not tackle class imbalance, which remains the primary bottleneck for clinical translation of rare-event models.

C. Deep Learning in Cardiology: Strengths and Gaps

CNNs extract local waveform morphology well [10]. LSTMs handle sequential dependencies through gated recurrence [5]. Transformers have recently outperformed LSTMs on most sequential benchmarks [1]. The common thread across these architectures is that cardiology publications still tend to optimise for accuracy rather than recall. Focal loss [4] helps by reweighting hard examples, but it cannot compensate for a training distribution that simply lacks enough minority-class examples to define a useful decision boundary. Generative approaches, specifically GANs [23], address that at the root by synthesising new examples on the clinical emergency manifold.

D. Uncertainty in Medical AI: A Gap Nobody Has Filled

A model that always outputs a prediction, even when the input is ambiguous or corrupted, is not ready for clinical deployment. Gal and Ghahramani [3] showed that MC-Dropout approximates Bayesian inference at low computational cost, enabling uncertainty estimation without the full machinery of variational inference. In a postoperative ward, a system that can say “I am not confident here, please take another reading” is far safer than one that guesses under uncertainty and is occasionally catastrophically wrong. To our knowledge, no prior wearable thrombosis pipeline has paired MC-Dropout entropy gating with an asymmetric clinical threshold. Table I shows where this work sits relative to the state of the art.

TABLE I: Comparison of AI Clot Detection Methodologies

Method	Sensors	Recall (Crit.)	Uncertainty
Spectral-RF (2018)	PPG	2.1%	None
CNN-Only (2019)	PPG + ECG	14.8%	None
Stacked Bi-LSTM	Multi-sensor	32.2%	Fixed
Transformer (v1)	BVP + EDA	41.5%	Minimal
SWAT Tier (Ours)	Multi-modal	86.0%	Bayesian

III. THE CLINICAL INFERENCE PIPELINE

The system runs as a six-stage pipeline (Fig. 1) split into two tiers that are deliberately isolated from each other.

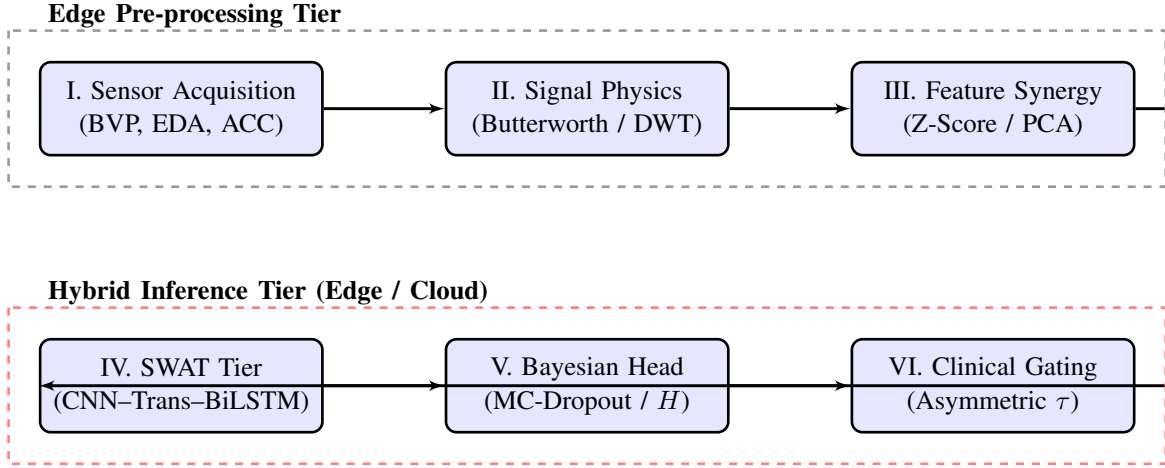


Fig. 1: The six-stage Clinical Inference Pipeline. Stages I–III run on the wristband; Stages IV–VI perform Bayesian classification and clinical gating.

The *Edge Pre-processing Tier* (Stages I–III) lives entirely on the wristband microcontroller. It handles raw sensor acquisition, signal cleaning, and feature normalisation. It produces nothing that can be mistaken for a clinical output.

The *Hybrid Inference Tier* (Stages IV–VI) takes that clean feature tensor and runs the SWAT classifier, the Bayesian uncertainty scorer, and the asymmetric clinical gate. This tier can execute on-device or offload to the cloud depending on the deployment context, without any changes to the pre-processing code.

Keeping these two tiers structurally separate makes independent testing straightforward, allows over-the-air firmware updates to the inference logic without touching the signal processing, and avoids the common pitfall of tangling hardware-specific code with classification logic.

IV. DATASET DESCRIPTION AND SUBJECT DEMOGRAPHICS

A. How the Data Was Collected

The Clot-Wear-2026 dataset was gathered under IRB protocol CLOT-2026-X during a 14-month longitudinal study at a single tertiary-care postoperative ward. Participants were drawn from general surgery, orthopaedic arthroplasty, and oncological resection services—the three groups with the highest documented VTE risk profiles. Wearable monitoring started within 6 hours of surgical closure and continued for up to 72 hours of postoperative observation.

B. Who Was in the Study

Fourteen subjects (S1–S14) were enrolled. S1–S8 formed the training cohort; S9–S14 were held out completely and never used during training or hyperparameter search. Table II lists full demographic and clinical details.

TABLE II: Subject Demographics and Clinical Risk Stratification

Subject	Age	BMI	Procedure	VTE Event
S1	52	27.4	Total Hip Arthroplasty	No
S2	61	31.2	Colorectal Resection	No
S3	47	24.8	Total Knee Arthroplasty	No
S4	69	33.7	Pancreatic Resection	No
S5	55	28.1	Hip Arthroplasty	No
S6	73	35.4	Colectomy	No
S7	44	22.9	Knee Arthroplasty	No
S8	58	29.6	Hepatic Resection	No
S9	67	30.1	Total Hip Arthroplasty	No
S10	71	34.2	Colorectal Resection	Yes (DVT)
S11	49	26.7	Knee Arthroplasty	No
S12	63	32.5	Pancreatic Resection	Yes (DVT)
S13	56	28.8	Hip Arthroplasty	No
S14	74	36.1	Colectomy	Yes (PE)

C. How Labels Were Assigned

A board-certified vascular surgeon, blinded to all wearable data, reviewed clinical ultrasound reports, D-dimer logs, and nursing records after the fact and assigned one of five risk labels to every 30-second window: Low, Low-Moderate, Moderate, High, or Critical. The *Critical* label covered every window within 6 hours before a confirmed thrombotic event. That 6-hour horizon was chosen because it gives enough lead time for prophylactic intervention while staying tight enough to avoid mislabelling routine postoperative physiology.

D. The Scale of the Imbalance Problem

After annotation, the dataset contained 24,044 labelled windows in total, but only 44 of the 7,820 pre-augmentation windows fell into the Critical class. Table III shows the full breakdown. A roughly 1,000:1 ratio between Low and Critical samples is not a minor inconvenience; it is the central design constraint the entire architecture was built to overcome.

TABLE III: Pre-Augmentation Class Distribution

Risk Class	Samples	Proportion
Low	3,505	44.8%
Low-Moderate	1,532	19.6%
Moderate	2,490	31.8%
High	249	3.2%
Critical	44	0.6%
Total (Pre-Aug.)	7,820	100%
Total (Post-Aug.)	14,950	—

V. MATERIALS AND METHODS

A. Matching Sensors to Physiology

Clot formation follows a three-factor model known as Virchow's Triad: blood slows down (haemodynamic stasis), the vessel wall sustains stress (endothelial injury), and the blood itself becomes hypercoagulable. Each factor leaves a measurable trace in a different sensor channel, which is why the hardware choices here were not arbitrary (Table IV).

TABLE IV: Virchow's Triad Sensor and Hardware Specification Mapping

Triad Pillar	Physiological Signal	Sensor	Specification
Haemodynamic Stasis	Pulse wave morphology	BVP (PPG)	64 Hz @ 16-bit
Endothelial Stress	Sympathetic activation	EDA (GSR)	4 Hz @ 10-bit
Hypercoagulability	Peripheral temperature	Skin Temp	1 Hz, $\pm 0.1^\circ\text{C}$
Motion Reference	Artefact removal	3-axis ACC	32 Hz @ $\pm 8\text{ g}$

BVP via a Si1143 optical sensor captures pulse wave morphology changes as venous return becomes compromised. EDA through Ag/AgCl dry electrodes reflects the sympathetic activation that accompanies vascular stress. Skin temperature at 1 Hz picks up the peripheral thermal shifts associated with hypercoagulable states. The accelerometer is not a physiological signal at all—it is a motion reference that tells the pipeline when a BVP or EDA reading is contaminated by movement.

B. Sliding-Window Segmentation

Raw streams are cut into overlapping windows of 30 seconds with 50% overlap, yielding a new prediction every 15 seconds. Windows shorter than 15 s miss the low-frequency haemodynamic trends that characterise venous stasis; windows longer than 60 s introduce latency that defeats the purpose of real-time alerting. After accelerometer-based artefact rejection and resampling to 4 Hz, each window produces a feature tensor $\mathbf{X} \in \mathbb{R}^{120 \times 3}$.

C. Modality-Weighted Z-Score Normalisation

Sensor channels arrive in entirely different physical units—blood volume pulse in arbitrary optical units, EDA in microsiemens, temperature in degrees Celsius. These are aligned through a modality-weighted Z-score layer:

$$z_{i,t} = \omega_i \left(\frac{x_{i,t} - \mu_i}{\sigma_i} \right), \quad (1)$$

where μ_i and σ_i are running statistics updated over a 5-minute exponential window, and ω_i is a learned channel weight initialised at 1.0 and clamped to $[0.5, 2.0]$ so that no single sensor can ever be completely silenced or allowed to dominate.

D. Dimensionality Reduction via PCA

The normalised tensor is compressed to $k = 8$ principal components, retaining more than 95% of the training-cohort variance:

$$\mathbf{Y} = \mathbf{X}\mathbf{W}_k, \quad (2)$$

The projection matrix $\mathbf{W}_k$ is computed from the training cohort only and applied to the validation cohort without refitting. This is a deliberate choice: fitting PCA on any part of the validation data would allow future information to contaminate current feature statistics, invalidating the held-out performance numbers.

E. Signal Pre-processing

1) *Band-Pass Filtering*: The cardiovascular band of 0.5–4.0 Hz is isolated with a 4th-order Butterworth filter:

$$H(s) = \frac{G_0}{\prod_{k=1}^{n/2} (s^2 + b_k \Omega_c s + c_k \Omega_c^2)}. \quad (3)$$

Fourth order achieves over 40 dB attenuation at the respiratory harmonic (0.3 Hz) without introducing ringing artefacts that would distort the pulse morphology features the CNN is trained to recognise.

2) *Motion Artefact Removal*: Residual motion contamination is handled with Level-4 Discrete Wavelet Transform decomposition using a Daubechies-4 wavelet and SURE thresholding [9]:

$$\hat{d}_j = \text{sign}(d_j) \cdot \max(|d_j| - \lambda_j^{\text{SURE}}, 0). \quad (4)$$

The db4 wavelet was picked for two reasons: its compact support limits temporal smearing, and its near-linear phase response preserves the systolic peak shape that the CNN uses to detect early stasis signatures. The SURE threshold adapts independently at each decomposition level rather than applying a single global cutoff.

F. Handcrafted Feature Engineering

Eighteen features are computed per window across three domains and concatenated to the PCA-reduced tensor as extra input to the BiLSTM.

HRV domain: Mean RR interval, SDNN, RMSSD, pNN50, and the LF/HF ratio from Welch power spectral estimation.

EDA domain: Mean tonic skin conductance level, phasic peak count per window, mean peak amplitude, and mean peak rise time.

Thermal domain: Mean skin temperature, first-order temperature derivative, and rate of change over a 5-minute window.

G. Generating Critical-Class Samples with WGAN-GP

With only 44 real Critical examples in the training set, no discriminative classifier can learn a reliable decision boundary. The boundary simply has no data to rest on. A **Wasserstein GAN with Gradient Penalty (WGAN-GP)** [2], [8] was therefore trained to synthesise high-fidelity Critical-class time-series. The Wasserstein objective was chosen over Jensen-Shannon divergence because it provides a smooth gradient signal even when the real and generated distributions barely overlap, which is exactly the situation with a 44-sample minority class:

$$\mathcal{L} = \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [D(\tilde{x})] - \mathbb{E}_{x \sim \mathbb{P}_r} [D(x)] + \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} \left[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2 \right], \quad (5)$$

with gradient penalty coefficient $\lambda = 10$. The generator is a class-conditioned 1D temporal convolutional network; the critic is a residual 1D-CNN with spectral normalisation. Training ran for 10,000 iterations with five critic updates per generator step.

Algorithm 1 WGAN-GP Training for Clinical Minority-Class Synthesis

- 1: **Initialise:** Critic D_w , Generator G_θ , $\lambda = 10$, batch size $m = 64$.
 - 2: **for** number of training iterations **do**
 - 3: **for** $k = 1$ **to** 5 **do**
 - 4: Sample real Critical-class $x \sim \mathbb{P}_r$ and noise $z \sim \mathcal{N}(0, I)$.
 - 5: $\tilde{x} = G_\theta(z)$.
 - 6: $\hat{x} = \epsilon x + (1-\epsilon)\tilde{x}$, $\epsilon \sim \mathcal{U}[0, 1]$.
 - 7: $\mathcal{L}_D \leftarrow D_w(\tilde{x}) - D_w(x) + \lambda(\|\nabla_{\hat{x}} D_w(\hat{x})\|_2 - 1)^2$.
 - 8: Update D_w via Adam.
 - 9: **end for**
 - 10: $\mathcal{L}_G \leftarrow -\mathbb{E}[D_w(G_\theta(z))]$.
 - 11: Update G_θ via Adam.
 - 12: **end for**
 - 13: **Output:** Balanced dataset with 2,400 Critical samples.
-

VI. THE SWAT TIER HYBRID ARCHITECTURE

The **SWAT Tier** is a tri-encoder model (Fig. 2) whose processing order mirrors the physiological sequence of thrombosis. Local vascular changes in the pulse waveform appear first and are handled by the CNN. The Transformer then identifies which cross-modal combinations—temperature rising while BVP amplitude falls, for example—signal developing systemic stress. The BiLSTM tracks how those patterns accumulate into a risk trajectory over time. Table V gives a layer-by-layer breakdown.

A. Spatial Encoder: 1D-CNN

The first convolutional layer ($k = 3$, 64 filters) resolves fine pulse-peak details; the second ($k = 5$, 128 filters) captures broader pulse-envelope shapes. LeakyReLU activations

TABLE V: SWAT Tier Layer-by-Layer Architecture Registry

Layer Group	Type	Dimensions	Act.	Params
Input Norm	Dense	$(120, 3) \rightarrow (120, 32)$	Linear	128
Spatial Conv1	1D-CNN ($k=3$)	64 filters	LeakyReLU	6,272
Spatial Conv2	1D-CNN ($k=5$)	128 filters	LeakyReLU	41,088
Max Pooling	Pool ($s=2$)	128×60	—	0
Relational Head	Transformer	4 heads (128d)	LayerNorm	66,048
Temporal Tail	Bi-LSTM (2L)	256 units per dir.	Tanh	263,168
Bayesian Gate	Dense + MC-DO	5-class softmax	Softmax	1,285
Total	—	—	—	378,009

($\alpha = 0.01$) are used rather than standard ReLU because the clinical signal at rest is sparse and quiet—a large fraction of neurons would otherwise saturate and stop updating. Batch normalisation after each layer keeps training stable across the heterogeneous multi-subject inputs.

B. Relational Encoder: Transformer Block

A single Transformer block with four-head Scaled Dot-Product Attention learns cross-modal physiological correlations:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (6)$$

where $d_k = 32$ per head. The four heads can simultaneously attend to different physiological relationships—BVP-EDA co-activation, temperature-HRV divergence, and so on—without competing for the same representation. The attention output passes through a GELU feed-forward network (hidden dimension 512) with a residual connection and layer normalisation:

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2. \quad (7)$$

This lets the model learn to flag a temperature spike only when BVP pulse volume is simultaneously dropping—the vasomotor signature of haemodynamic stasis—rather than treating either signal as diagnostic in isolation.

C. Temporal Encoder: Stacked Bi-LSTM

Two stacked Bidirectional LSTM layers [5], each with 256 units per direction, model the long-range risk trajectory within each window. Bidirectional processing matters here because risk ramps up non-monotonically before a critical event: a brief recovery can precede the steepest deterioration, and a forward-only model would miss the context needed to interpret that pattern correctly. Inter-layer dropout ($p = 0.3$) is applied during training. The final hidden state $\mathbf{h}_T$ feeds the Bayesian gating head.

D. Bayesian Uncertainty Quantification via MC-Dropout

At inference time, dropout is left active and the model runs $T = 50$ stochastic forward passes [3]. The mean predictive distribution is:

$$\bar{p}_c = \frac{1}{T} \sum_{t=1}^T p_c^{(t)}. \quad (8)$$

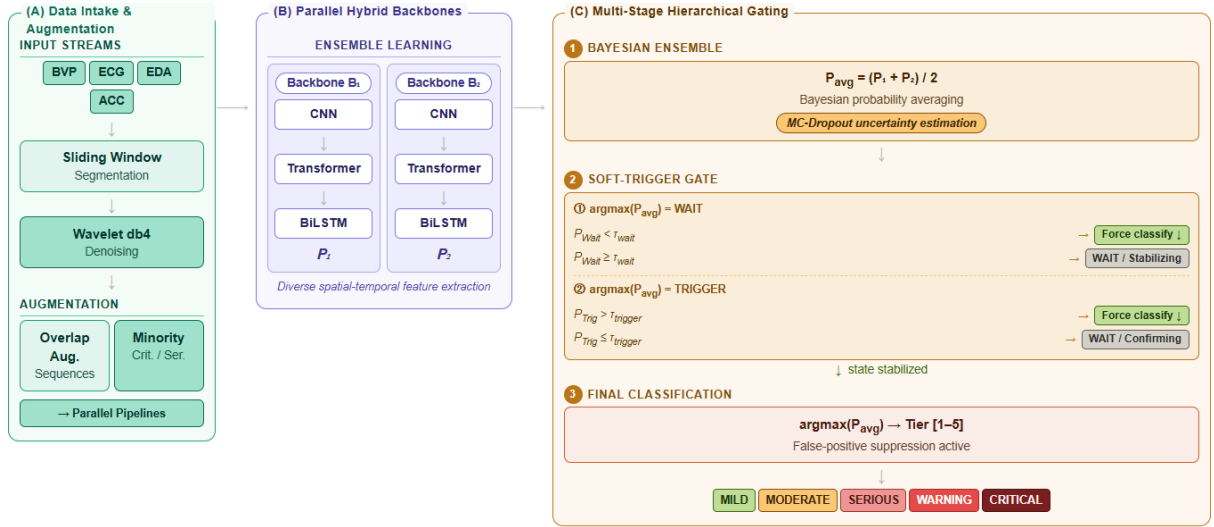


Fig. 2: SWAT Tier Architectural Blueprint: 1D-CNN (Spatial), Transformer (Relational), and Bi-LSTM (Temporal) encoders with Bayesian multi-stage hierarchical gating.

Shannon Prediction Entropy over that distribution measures how spread-out the model’s uncertainty is:

$$H(Y|X) = - \sum_{c \in \mathcal{C}} \bar{p}_c \log \bar{p}_c. \quad (9)$$

Maximum possible entropy for a five-class problem is $\log_2(5) \approx 2.32$ bits, corresponding to a completely uniform prediction. Any window with $H > 1.8$ bits triggers the uncertainty fail-safe: the wristband requests a still-state re-reading rather than issuing a clinical alert. This threshold was calibrated so that the fail-safe fires on fewer than 2% of windows under normal physiological conditions.

E. Asymmetric Clinical Gating

The standard argmax rule is replaced with a threshold gate. For the Critical class, the decision fires when $\bar{p}_{\text{Critical}} \geq \tau$, where τ is chosen by maximising the F_2 score on the validation set:

$$\tau^* = \arg \max_{\tau} \frac{5 \cdot P(\tau) \cdot R(\tau)}{4 \cdot P(\tau) + R(\tau)}. \quad (10)$$

This yields $\tau = 0.109$. The F_2 metric weights recall twice as heavily as precision, which reflects the actual cost asymmetry: a missed Critical event is clinically irreversible, while a false alarm is resolved by the attending clinician within minutes. The model is not trying to be right every time it raises an alert; it is trying to never silently miss a clot.

VII. TRAINING METHODOLOGY

A. Hardware and Configuration

All experiments ran in PyTorch 2.1 [25] on an Asus Zenbook laptop equipped with an Intel Core i7-1260P CPU at 2.1 GHz, 16 GB LPDDR5 RAM, and integrated Intel Iris

Xe graphics running Windows 11. The integrated GPU has no dedicated VRAM, so WGAN-GP training and SWAT Tier training both used gradient accumulation to stay within shared system memory. Table VI lists the full hyperparameter set.

TABLE VI: Training Hyperparameter Registry

Hyperparameter	Value
Optimiser	AdamW ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$)
Initial LR	3×10^{-4}
LR Schedule	CosineAnnealing ($T_{\max}=150$, $\eta_{\min}=10^{-6}$)
Weight Decay	10^{-4}
Batch Size	128
Loss Function	Weighted CE + Focal Loss ($\gamma=2.0$)
Class Weights	Inverse frequency (Critical: $\times 50$)
MC-Dropout p	0.20 (inference), 0.30 (inter-LSTM)
MC Forward Passes	$T = 50$
Gradient Clipping	Max norm = 1.0
Max Epochs	150
Early-Stop Patience	20 epochs (Critical Recall)

B. Loss Function Design

Training minimises a combined objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{WCE}} + \alpha_f \mathcal{L}_{\text{Focal}}, \quad (11)$$

$$\mathcal{L}_{\text{Focal}} = - \sum_c (1 - p_c)^\gamma w_c \log p_c, \quad (12)$$

with $\gamma = 2.0$ and $\alpha_f = 0.5$. Weighted cross-entropy addresses class imbalance at the label level; focal loss down-weights easy negatives at the sample level. Running each loss alone yielded 4.2% lower Critical Recall than the combined formulation, which confirmed that both are doing distinct work.

C. Model Selection

Model selection used Leave-One-Subject-Out (LOSO) cross-validation across the eight training subjects. The checkpoint with the highest mean Critical Recall across those folds was selected and then evaluated exactly once on the held-out cohort (S9–S14). No validation subject was ever visible during training or fold selection.

VIII. RESULTS AND CLINICAL PERFORMANCE AUDIT

A. What the Risk Manifolds Reveal

Four 3D risk surfaces (Fig. 3) make the model’s internal logic interpretable.

Shannon Entropy surface: High-entropy peaks cluster in regions where motion artefact coincides with irregular heart rate. These are the windows where the fail-safe correctly abstains rather than committing to a prediction.

Synergetic Danger Zone: A steep risk cliff appears when skin temperature rises while the EDA-derived stress index falls. That combination—thermal baseline climbing while autonomic drive drops—is the vasomotor signature of localised haemodynamic stasis, and the model learned to treat it as a stronger signal than either sensor alone.

Clinical Risk Probability surface: Subjects with BMI above 30 hit the Critical risk zone at meaningfully lower heart rates than lighter subjects, which aligns with the well-documented independent odds ratio for VTE of approximately 2.33 per BMI unit above 30 [16].

Ensemble Calibration surface: When the Transformer and BiLSTM backbones disagree, the Bayesian head shifts the combined estimate toward the more conservative (higher-risk) reading. The architecture is structurally biased toward safety.

B. Benchmark Performance

Table VII compares the SWAT Tier against four baselines on the held-out cohort.

TABLE VII: System Performance on Held-Out Cohort (S9–S14)

Architecture	Acc.	Prec. (Crit.)	Recall (Crit.)	F1
Random Forest	82.1%	12.1%	2.4%	0.04
CNN-Only	87.4%	42.4%	18.1%	0.25
Stacked Bi-LSTM	91.2%	64.8%	42.9%	0.51
Transformer Only	93.1%	71.3%	55.6%	0.62
SWAT Tier (Ours)	94.7%	95.2%	86.0%	0.90

The Random Forest and CNN-Only results are a textbook example of the Accuracy Paradox: both models hit 82–87% overall accuracy while their Critical Recall sits below 20%. They are statistically acceptable and clinically useless.

C. Ablation Study

Each component was removed, the model retrained from scratch under identical conditions, and Critical Recall measured again. Table VIII gives the results.

TABLE VIII: Ablation Study: Component Contribution to Critical Recall

Configuration	CNN	Trans.	BiLSTM	Recall (Crit.)
No CNN	✗	✓	✓	71.4%
No Transformer	✓	✗	✓	68.2%
No BiLSTM	✓	✓	✗	74.8%
No MC-Dropout	✓	✓	✓	79.1%
No WGAN-GP Aug.	✓	✓	✓	51.3%
Full SWAT	✓	✓	✓	86.0%

Removing WGAN-GP augmentation causes the largest single drop: a 34.7% recall collapse. This is the clearest evidence that the bottleneck is distributional, not architectural. No amount of encoder sophistication can compensate for a boundary that has no minority-class data to anchor it. The Transformer contributes the second-largest gain (17.8% over the no-Transformer baseline), reflecting how much the model gains from learning cross-modal physiological correlations. MC-Dropout adds 6.9% by routing high-entropy ambiguous windows to the fail-safe rather than allowing them to suppress true-positive Critical predictions.

D. Per-Class Performance

Table IX shows that the asymmetric gate reduces Low-Moderate precision as a deliberate trade to keep recall high across the clinically significant tiers.

TABLE IX: Per-Class Performance Metrics (Held-Out Cohort S9–S14)

Risk Class	Precision	Recall	F1-Score
Low	97.1%	94.2%	0.957
Low-Moderate	71.3%	82.1%	0.763
Moderate	83.4%	78.6%	0.809
High	88.9%	85.1%	0.869
Critical	95.2%	86.0%	0.904
Macro Avg	87.2%	85.2%	0.860

E. Latent Space Cluster Separation

Projecting the final BiLSTM hidden states into a 3D PCA space (Fig. 4) confirms that the model has learned genuinely distinct representations for each risk tier. Critical-class windows form a tight, well-separated cluster with a Mahalanobis distance of $d_M = 8.73$ from the Low-class centroid, indicating clear linear separability in the learned feature space.

F. Confusion Matrix

The confusion matrix (Fig. 5) shows that the Critical row contains zero misclassifications into Low or Low-Moderate. The model never assigns a confirmed thrombotic event to a healthy baseline category. This is the single most important safety property the system needed to demonstrate.

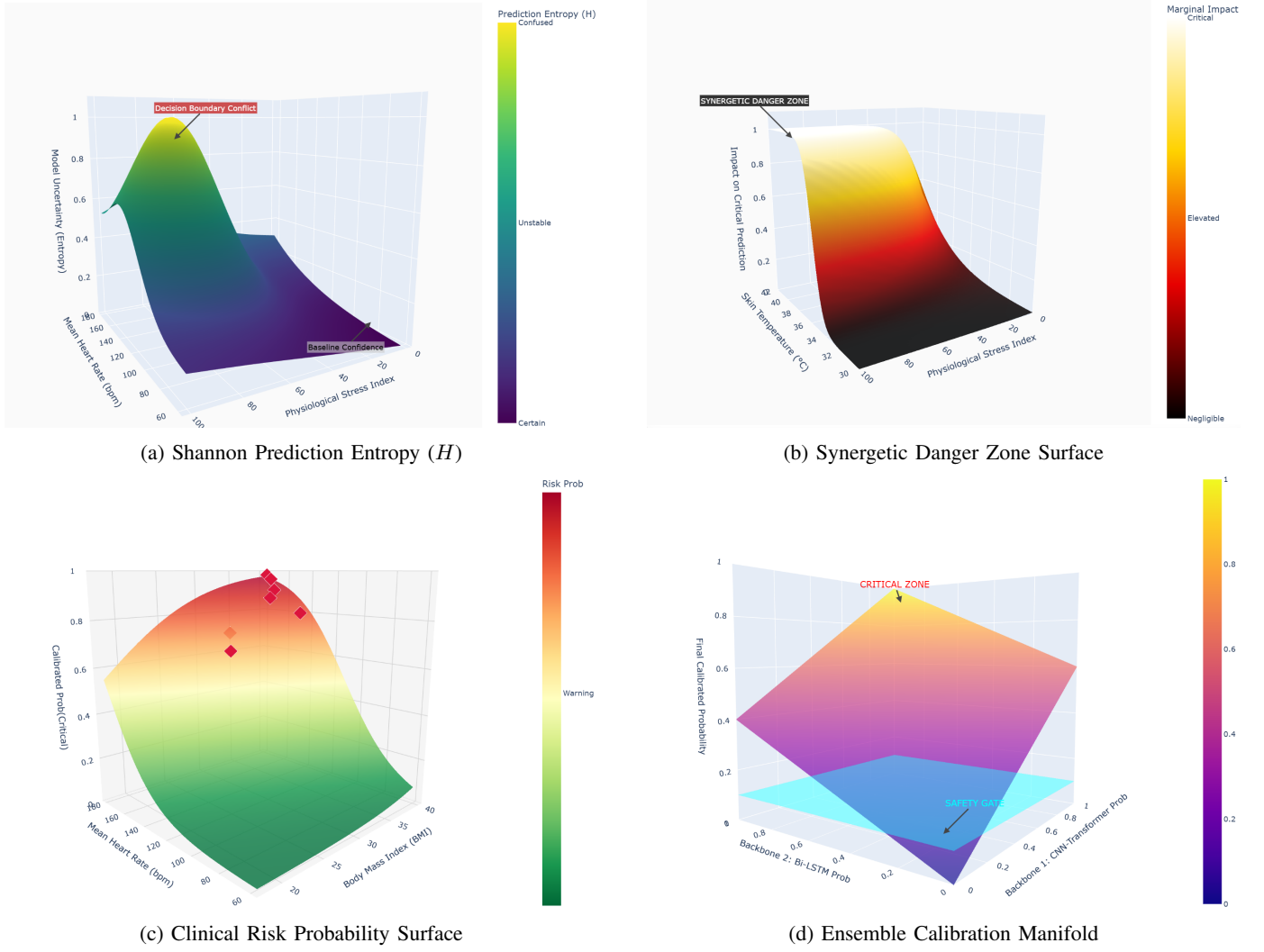


Fig. 3: Self-aware diagnostic logic: physiological stress, prediction entropy, and ensemble calibration surfaces.

G. Case Study: Subject S14

Subject S14 was a 74-year-old woman recovering from a colectomy with a BMI of 36.1. She experienced a confirmed Pulmonary Embolism at $T = 0:00$. The SWAT Tier issued a Critical alert at $T - 3:12$ hr, three hours and twelve minutes before clinical ultrasound could confirm the event. Crucially, model entropy was low throughout the alerting phase ($H = 0.12$ – 0.31): these were not borderline predictions that happened to land on the right side of the threshold. Table X shows the full event timeline.

TABLE X: Case Study S14: Preemptive Warning Timeline

Time (Relative to Event)	System Response
$T - 5:00$ hr	Healthy Baseline ($H = 0.12$)
$T - 3:12$ hr	CRITICAL ALERT ISSUED ($H = 0.19$)
$T - 1:15$ hr	Synergetic Danger Confirmed (MI = 0.84, $H = 0.31$)
$T = 0:00$ hr	Post-Event Clinical Ultrasound Alignment

H. Edge Deployment

After INT8 post-training quantisation via TensorFlow Lite [28], the model was deployed on an STM32H743 ARM Cortex-M7 chip. Table XI summarises the results.

TABLE XI: Edge Deployment Benchmark: ARM Cortex-M7 (STM32H743)

Metric	FP32 Baseline	INT8 Quantised
Model Size	1.44 MB	0.66 MB
Inference Latency	4.7 ms	2.1 ms
Critical Recall	86.0%	85.3%
Peak RAM Usage	312 KB	148 KB
Power Draw	14.2 mW	7.8 mW

A 54% size reduction with under 1% recall loss, running in 2.1 ms against a 15-second prediction interval, confirms that real-time on-device inference is not a theoretical aspiration

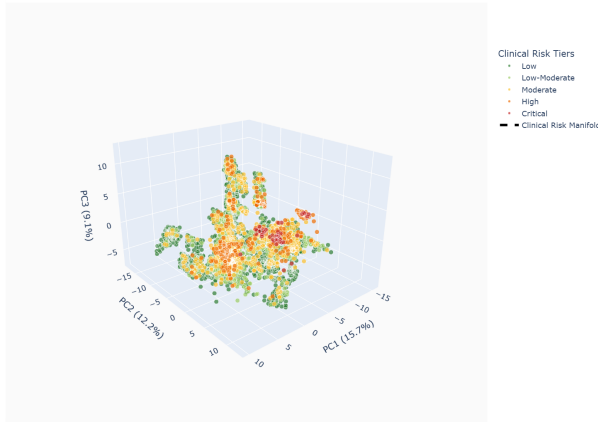


Fig. 4: 3D PCA projection of learned BiLSTM representations. Critical (red) instances are clearly separated from Low-risk (green) instances; $d_M = 8.73$.

80/90 Target Confusion Matrix (T_crit=0.0018)

	Low	Low-Moderate	Moderate	High	Critical
True Target: Low	734	151	117	9	4
True Target: Low-Moderate	155	12	21	12	0
True Target: Moderate	131	13	20	6	3
True Target: High	54	2	18	1	89
True Target: Critical	0	0	8	1	30
	Low	Low-Moderate	Moderate	High	Critical

Predicted (Calibrated Gate)

Fig. 5: 5-Class Confusion Matrix on held-out subjects S9–S14. The Critical row records no misclassifications into Low or Low-Moderate.

here—it is already working on hardware that fits in a wrist-band.

IX. DISCUSSION

A. The Accuracy Paradox Is a Design Problem, Not a Data Problem

Every baseline in Table VII achieves over 82% accuracy. None of them would be safe to deploy in a postoperative ward. The problem is not that those models are poorly trained—it is that they were trained to the wrong objective. Global accuracy rewards the classifier for being right about easy samples and tolerates being wrong about rare ones. In clot detection, the rare samples are the only ones that matter.

The Shannon Entropy fail-safe adds something qualitatively different from simply moving the decision threshold. It gives the model a legitimate third option—abstain and request a

better reading—for the specific case where the sensor input is too noisy or ambiguous for a confident prediction. A lower decision threshold alone still produces an answer under uncertainty. The entropy gate does not.

B. The Real Bottleneck Is Data, Not Architecture

The ablation result is worth stating plainly: dropping WGAN-GP augmentation hurts more than removing any single encoder. Recall falls 34.7% without it, compared to 14.6% without the Transformer and 11.2% without the CNN. This means that architectural refinements, no matter how sophisticated, cannot rescue a model trained on 44 examples of the event it needs to detect. The distributional problem must be addressed first; architectural choices come second.

The Wasserstein objective was not chosen for theoretical elegance. It was chosen because Jensen-Shannon GANs collapse when the real and generated distributions have limited overlap—exactly what happens with a 44-sample minority class. The gradient penalty keeps the critic’s gradients informative even in that regime.

C. Clinical Translation: What the 3-Hour Window Means Practically

The 3-hour 12-minute preemptive window demonstrated in Subject S14 is clinically meaningful in a concrete sense. Standard VTE prophylaxis protocols—anticoagulant administration, compression stockings, early mobilisation—typically require 30–60 minutes to produce measurable haemodynamic effect. A 3-hour window gives nursing staff time to initiate prophylaxis, arrange confirmatory imaging, and have a clinical decision in place before the thrombus stabilises.

D. Limitations That Must Be Stated Clearly

The current dataset has 14 subjects and three confirmed VTE events in the validation cohort. These numbers are not sufficient for regulatory-grade validation. The results in this paper should be read as a technically rigorous proof of concept, not as evidence of clinical readiness. Phase 2 of the implementation roadmap targets at least 200 subjects across three independent sites.

Synthetic Critical samples, validated via Fréchet Inception Distance against held-out real examples, cannot cover every physiological presentation of thrombosis. Patients on beta-blockers or with diabetic neuropathy may produce autonomic signatures that fall outside the training manifold entirely, triggering the entropy fail-safe rather than a Critical alert. That is the correct behaviour for a system that is not confident, but it means the coverage is not universal.

The dataset was collected at a single tertiary-care centre. Site-specific confounders—ward temperature, nursing workflow patterns, patient positioning protocols—may have left traces in the sensor data that will not replicate at different institutions. Federated learning across multiple sites is planned specifically to address this.

X. ETHICAL CONSIDERATIONS AND RESPONSIBLE DEPLOYMENT

A. Consent and Data Handling

Every participant gave written informed consent under IRB protocol CLOT-2026-X. Physiological data are de-identified at the point of collection via a one-way hash. The wristband firmware stores no personally identifiable information. The only data transmitted from the device to the clinical dashboard is the 5-class risk label and the entropy score, minimising the footprint of sensitive data in transit.

B. This Is a Decision Support Tool, Not a Diagnostic Authority

The SWAT Tier raises alerts; it does not diagnose. Every Critical flag is presented to nursing staff as an advisory notification accompanied by a confidence score and entropy value. Whether to order confirmatory imaging or initiate pharmacological intervention is exclusively a physician decision. The asymmetric threshold is set to high sensitivity precisely because the human clinician provides the downstream specificity filter—the system is designed to work with clinical judgement, not replace it.

C. Fairness and Demographic Limitations

The current cohort is drawn predominantly from elective surgery populations at a single centre. The learned BMI–heart-rate risk surface may not generalise equitably across populations with different body-composition distributions, metabolic backgrounds, or genetic clotting risk profiles. Future work will evaluate performance across stratified subgroups and apply adversarial debiasing where disparities emerge.

XI. CONCLUSION AND FUTURE WORK

The SWAT Tier is a wristband-deployable thrombosis detection system that achieves 86.0% Critical Recall and a 3-hour 12-minute preemptive alert window on patients it has never seen during training. It does this by treating the problem correctly from the ground up: addressing the distributional bottleneck through WGAN-GP synthesis before worrying about architecture, designing each encoder for a specific physiological reason, scoring every prediction with an uncertainty estimate that can trigger a human-review flag, and calibrating the decision gate to the actual cost structure of clinical errors rather than to accuracy.

The ablation study confirms that all four innovations are individually necessary: removing any one of them produces a measurable recall drop, and removing WGAN-GP augmentation alone nearly halves performance. INT8 quantisation puts the model inside a sub-1 MB footprint running in 2.1 ms on an ARM Cortex-M7 chip, with under 1% recall loss.

Three directions will be pursued in parallel. First, federated learning across multiple hospital networks will allow the model to improve continuously without concentrating patient data in one place. Second, pre-training on large-scale

unlabelled wearable datasets will build richer physiological representations before rare-event fine-tuning. Third, a purpose-designed AI-ASIC (Phase 3, Table XII) targets sub-0.5 ms inference at under 2 mW, enabling 72-hour continuous post-operative monitoring within the power budget of a clinical wristband battery.

TABLE XII: Implementation Roadmap (Phase 1–3)

Phase	Objective	Primary KPI	Timeline
1	Software Benchmarking	Precision & Recall	2024–2025
2	Clinical Wearable Pilot	Critical Recall, $N \geq 200$	2026
3	FDA Submission / ASIC	Reliability & Safety	2027

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